

Skills or Parks?

A Nested Test of Dynamic Complementarity versus Preference Formation in Early Childhood*

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Abstract

The technology of skill formation (Cunha and Heckman, 2007; Cunha et al., 2010) and the adaptive preference formation model both explain why early childhood interventions produce persistent non-cognitive gains. They disagree on the mechanism: skill complementarity versus preference deposits. I show the two models are nested by a single parameter—the cross-partial of the transition function with respect to current traits and subsequent investment. Under skills, this cross-partial is positive (dynamic complementarity: the treatment raises the return to post-treatment investment). Under formation, it is zero (the preference deposit decays at a cultural persistence rate independent of the post-treatment environment). Existing quasi-experimental evidence on dynamic complementarity is genuinely mixed: Johnson and Jackson (2019) find that Head Start and subsequent school spending interact positively, while Goff et al. (2025) find substitutability in Romanian data and Schiman (2024) finds no complementarities in the Head Start Impact Study. I derive five discriminating predictions from the nested model, map each to specific published tables and figures from the Perry, Abecedarian, Head Start, and Jamaica experiments, and show that the joint pattern of evidence—including the critical distinction between cognitive and non-cognitive trajectories—favors the formation model for the non-cognitive margin and the skills model for the cognitive margin. The two mechanisms coexist in the same children, operating on different dimensions of development.

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1 Introduction

Why do early childhood interventions produce persistent returns? Two frameworks offer structurally distinct answers.

The technology of skill formation (Cunha and Heckman, 2007; Cunha et al., 2010) models non-cognitive traits as skills produced by a CES technology with dynamic complementarity: early skills raise the marginal product of subsequent investment. Persistence arises because skills beget skills—the treated child’s elevated skill stock makes every subsequent year of schooling more productive, compounding the initial advantage. The structural estimates of Cunha et al. (2010), using CNLSY/79 data on 2,207 firstborn white children, find that the complementarity parameter ϕ for noncognitive skills is strongly negative in early childhood ($\hat{\phi} = -0.852$, s.e. 0.199; Table 4, p. 914), implying an elasticity of substitution of only 0.54—substantial complementarity. For cognitive skills, complementarity *strengthens* with age ($\hat{\phi}$ falls from 0.331 to -0.797), while for noncognitive skills it *weakens* ($\hat{\phi}$ rises from -0.852 to -0.308). This asymmetry—noncognitive skills are more complementary early, less complementary late—is the engine of the Heckman curve.

The adaptive preference formation model models non-cognitive traits as preferences deposited by the developmental signal environment. Persistence arises from cultural transmission (ρ): the deposit decays slowly because intergenerational inertia, Bayesian posterior precision, and identity incorporation all resist preference revision. The age gradient in returns follows from the closure of the formation window: early interventions change the developmental environment during the years when preferences are still being written; late interventions try to overwrite a deposit that has already hardened.

Both frameworks are consistent with the canonical facts: large persistent non-cognitive returns (Schweinhart et al., 2005; García et al., 2020), cognitive fadeout (Bailey et al., 2017; Protzko, 2015), and a steep age gradient (Heckman and Mosso, 2014). Both generate the Heckman curve. Both rationalize benefit-cost ratios of 7–13 (Heckman et al., 2010; García et al., 2020). The question is whether they can be distinguished empirically.

This paper shows they can. The key insight is that the two models differ in one structural feature: the *cross-partial derivative* of the transition function with respect to current traits and subsequent investment. Under skill formation, this cross-partial is positive—dynamic complementarity means that a child with higher traits benefits more from subsequent investment. Under preference formation, this cross-partial is zero—the preference deposit decays at rate ρ regardless of the post-treatment environment, because both treated and control children face the same subsequent signals.

This generates a clean empirical test: does the treatment effect on non-cognitive out-

comes interact with the quality of the post-treatment environment? If yes: skills dominate. If no: formation dominates.

The existing quasi-experimental literature on dynamic complementarity sends mixed signals. Johnson and Jackson (2019), exploiting county-level variation in Head Start rollout and court-ordered school finance reforms in PSID data, find that Head Start’s long-run effects on educational attainment and wages are significantly larger in counties that subsequently received higher K–12 spending—strong evidence for $\delta > 0$. But Goff et al. (2025), using Romanian data combining regression discontinuity and differences-in-differences, find “no consistent evidence of interactions” between family environment and school quality, with patterns “suggestive of substitutability rather than complementarity.” Rossin-Slater and Wüst (2020), studying Danish preschool and nurse home visiting programs, find that prior nurse exposure *reduces* the added value of preschool by 80–88%—programs as partial substitutes, not complements. And Schiman (2024), directly testing complementarity in the Head Start Impact Study RCT, finds that “effects of preschool depreciated significantly within a year and that there were no positive dynamic complementarities that offset depreciation.”

This paper contributes three things. First, a structural nesting: a single transition function that contains both skill formation and preference formation as special cases, distinguished by one parameter δ . Second, five discriminating predictions that exploit the structural differences between the models, each mapped to specific published tables and figures. Third, a reconciliation: the evidence suggests that cognitive and non-cognitive dimensions operate through different mechanisms—cognitive outcomes follow the skills model (flow-dependent, interacting with post-treatment quality), while non-cognitive outcomes follow the formation model (deposit-dependent, invariant to post-treatment quality). The two mechanisms coexist in the same children.

This dual-process finding matters for policy. If noncognitive persistence operates through preference deposits ($\delta = 0$), the optimal early childhood policy concentrates resources on changing the signal environment during the formation window—the teachers, the role models, the visible futures—rather than spreading investment over 18 years to exploit complementarity. The curriculum fades. The park persists.

Related literature. This paper connects three strands. The first is the technology of skill formation (Cunha and Heckman, 2007; Cunha et al., 2010; Heckman and Mosso, 2014; Agostinelli and Wiswall, 2025) and its empirical tests (Johnson and Jackson, 2019; Goff et al., 2025; Rossin-Slater and Wüst, 2020; Schiman, 2024). The second is the economics of endogenous preferences. Bowles (1998) argued that markets and economic institutions

shape the preferences of the agents who participate in them—the premise of the formation model. Becker and Mulligan (1997) formalized the endogenous determination of time preference, showing that wealthier agents invest more in patience because the return to patience is higher. Their model predicts that transfers change patience (because transfers change wealth); the formation model predicts they do not (because transfers do not change the return signal). Bisin and Verdier (2001, 2023) developed the theory of cultural transmission of preferences, providing the microfoundation for ρ . Doepke and Zilibotti (2017) modeled parenting styles as endogenous choices for transmitting patience and other preferences—an explicit preference-formation mechanism operating through exactly the parental channel this paper’s Bisin-Verdier microfoundation describes. The third strand is the experimental evidence on preference malleability. Cappelen et al. (2020) found that a preschool program changed children’s social preferences (egalitarianism), with effects persisting 3.5 years after treatment—direct evidence that early childhood programs change preferences, not just skills. Castillo et al. (2024) found that the same experimental platform did *not* change time preferences, suggesting that different preference dimensions may have different formation mechanisms. Sen (1985) identified the phenomenon of adaptive preferences—the deprived adjusting their desires to match their circumstances—which the formation model formalizes as the signal-dependent updating of θ . This paper provides the first structural test that distinguishes the skill and preference channels of persistence within the same RCT framework.

2 Two Models of Persistence

The two frameworks agree that early childhood interventions produce persistent non-cognitive gains. They disagree on *why* the gains persist. This section derives each model’s persistence mechanism formally, then identifies the structural feature on which they diverge.

2.1 The Skill Formation Model

Cunha and Heckman (2007) introduced the multistage technology of skill formation. The CES specification (their Equation 6, p. 38) gives the transition from the current noncognitive stock θ_t and current investment I_t to the next-period stock:

$$\theta_{t+1} = f(\theta_t, I_t) = \left[\gamma \theta_t^\phi + (1 - \gamma) I_t^\phi \right]^{1/\phi}, \quad \phi \leq 1, \quad \gamma \in (0, 1). \quad (1)$$

The elasticity of substitution between the current stock and investment is $\sigma = 1/(1 - \phi)$. When $\phi = 1$, the two inputs are perfect substitutes ($\sigma = \infty$): investment can fully compensate for a low stock. When $\phi \rightarrow -\infty$, they are Leontief complements ($\sigma \rightarrow 0$): investment is useless without a prior stock. The intermediate case $\phi = 0$ gives Cobb-Douglas ($\sigma = 1$).

Two properties of f drive the model's predictions (pp. 35–37 of Cunha and Heckman 2007):

Self-productivity. The partial derivative with respect to the current stock is:

$$\frac{\partial f}{\partial \theta_t} = \gamma \theta_t^{\phi-1} \left[\gamma \theta_t^\phi + (1 - \gamma) I_t^\phi \right]^{1/\phi-1} > 0. \quad (2)$$

Skills persist on their own: a child with a higher stock today will have a higher stock tomorrow, holding investment constant. This is the *autoregressive* component of the technology. In isolation, self-productivity generates declining treatment effects: the effect decays at rate $\partial f / \partial \theta_t < 1$ (the function is concave in θ_t for $\phi < 1$), converging to zero in the absence of complementary investment.

Dynamic complementarity. The cross-partial derivative is:

$$\frac{\partial^2 f}{\partial \theta_t \partial I_t} = \gamma(1 - \gamma)(1 - \phi) \theta_t^{\phi-1} I_t^{\phi-1} \left[\gamma \theta_t^\phi + (1 - \gamma) I_t^\phi \right]^{1/\phi-2}. \quad (3)$$

This is strictly positive when $\phi < 1$: a child with a higher skill stock benefits more from a given unit of investment. Dynamic complementarity is the mechanism that makes early investment valuable: the preschool raises θ_0 , and the elevated stock amplifies the return to every subsequent year of schooling. Without complementarity ($\phi = 1$), the initial boost decays through self-productivity alone, and subsequent investment cannot arrest the decline.

What complementarity implies for treatment effects. Consider two children: treated (T , receiving $\Delta\theta_0 > 0$ from a preschool program) and control (C). Both enter the same

post-treatment school with investment I_1 . The treatment effect at $t = 1$ is:

$$D_1 = f(\bar{\theta} + \Delta\theta_0, I_1) - f(\bar{\theta}, I_1) \quad (4)$$

$$\approx \underbrace{\frac{\partial f}{\partial \theta} \Big|_{\bar{\theta}, I_1}}_{\text{self-productivity}} \Delta\theta_0 + \underbrace{\frac{1}{2} \frac{\partial^2 f}{\partial \theta^2} \Big|_{\bar{\theta}, I_1} (\Delta\theta_0)^2}_{\text{concavity correction}} + O(\Delta\theta_0^3).$$

The treatment effect depends on the investment level I_1 through the self-productivity term $\partial f / \partial \theta$, which is itself a function of I_1 . The *derivative* of the treatment effect with respect to post-treatment investment is:

$$\frac{\partial D_1}{\partial I_1} \approx \frac{\partial^2 f}{\partial \theta \partial I} \Big|_{\bar{\theta}, I_1} \Delta\theta_0 > 0 \quad \text{when } \phi < 1. \quad (5)$$

The treatment effect is *amplified* by higher post-treatment investment. A treated child in a good school benefits more from the treatment than a treated child in a bad school. This is the observable content of dynamic complementarity: the treatment $\times$ post-treatment-quality interaction should be positive.

Structural estimates. Cunha et al. (2010) estimated the CES technology using 2,207 first-born white children from the CNLSY/79. Their preferred specification (Table 4, p. 914, with unobserved heterogeneity) yields:

Dimension	Stage	$\hat{\phi}$ (s.e.)	$\hat{\sigma}$	Complementarity
Cognitive	Early	0.331 (0.107)	1.50	Substitutable
Cognitive	Late	−0.797 (0.077)	0.56	Strong complements
Noncognitive	Early	−0.852 (0.199)	0.54	Strong complements
Noncognitive	Late	−0.308 (0.077)	0.77	Moderate complements

Two features of these estimates matter for the nesting. First, noncognitive complementarity *weakens* with age (σ rises from 0.54 to 0.77), while cognitive complementarity *strengthens* (σ falls from 1.50 to 0.56). This means that by late childhood, investment complementarity is less important for noncognitive persistence than for cognitive persistence—the opposite of what a naive reading of the Heckman curve would suggest.

Second, noncognitive self-productivity *rises* with age: the share parameter on θ_t^N increases from 0.451 to 0.662 (Table 4). By later stages, 66% of next-period noncognitive skills come from the current stock perpetuating itself, versus only 34% from the combined

contribution of investment and cross-productivity. In the Cunha-Heckman-Schennach framework, *long-run noncognitive persistence is driven primarily by self-productivity, not by complementarity with investment*. The formation model offers a structurally different account of this self-productivity: the trait persists not because it raises its own marginal product but because it is a preference that was deposited and transmitted through cultural and cognitive channels.

An important methodological qualification: Agostinelli and Wiswall (2025) show that the mean-log-stationarity restriction used to identify the CES technology in Cunha et al. (2010) restricts the production function to the Cobb-Douglas sub-class ($\phi = 0$), potentially biasing complementarity estimates. The strong noncognitive complementarity ($\hat{\phi} = -0.852$) may partly reflect functional form rather than data.

2.2 The Formation Model

The preference formation model specifies the transition:

$$\theta_{t+1} = \rho \theta_t + (1 - \rho) \bar{\theta} + \mu(\lambda r_t - r^*), \quad (6)$$

where $\rho \in [0, 1)$ is cultural persistence, $\mu > 0$ is adaptation speed, $\lambda \in (0, 1]$ is the signal discount, r_t is the return signal from the environment, and r^* is the reference return.

This equation has three components: an autoregressive term ($\rho \theta_t$, how much of the current preference persists), a mean-reversion term ($(1 - \rho)\bar{\theta}$, the pull toward the population baseline), and an adaptation term ($\mu(\lambda r_t - r^*)$, how the current signal shifts the preference). The critical structural feature is that these three terms are *additive*. The autoregressive persistence of θ_t does not depend on the current signal r_t . The response to the current signal does not depend on the current level of θ_t .

Formally, the cross-partial is:

$$\frac{\partial^2 \theta_{t+1}}{\partial \theta_t \partial r_t} = 0. \quad (7)$$

This zero is the paper's structural content. Under the CES technology (1), the cross-partial is positive (equation 3). Under the formation model (6), it is exactly zero. The two models are distinguished by a single derivative.

The economic content of the zero. Why should the cross-partial be zero? The formation model offers a specific answer: the preference θ was formed by the *developmental* environment—the park—and persists through *cultural and cognitive* channels that are orthogonal to the current investment environment. A child whose park deposited high pa-

tience does not need a good school to remain patient. She is patient because twenty years of formative exposure built the patience into her worldview, and the worldview is transmitted intergenerationally through parenting, peer interaction, and identity formation—channels that operate regardless of school quality.

This is a substantive claim, not an algebraic convenience. It says that the mechanisms through which preferences persist—Bayesian posterior precision, cultural transmission, identity incorporation—are structurally different from the mechanisms through which skills compound. Skills compound through *production*: the skill stock raises the marginal product of subsequent input. Preferences persist through *transmission*: the preference is copied, inherited, or defended, independently of the external investment stream.

Treatment effects under formation. Consider the same treated and control children. The treated child received $\Delta\theta_0$ from the preschool program (the program changed her park). Both enter the same post-treatment environment with signal r_1 .

$$\begin{aligned}\theta_1^T &= \rho(\bar{\theta} + \Delta\theta_0) + (1 - \rho)\bar{\theta} + \mu(\lambda r_1 - r^*) = \bar{\theta} + \rho\Delta\theta_0 + \mu(\lambda r_1 - r^*), \\ \theta_1^C &= \rho\bar{\theta} + (1 - \rho)\bar{\theta} + \mu(\lambda r_1 - r^*) = \bar{\theta} + \mu(\lambda r_1 - r^*).\end{aligned}$$

The treatment effect at $t = 1$:

$$D_1 = \theta_1^T - \theta_1^C = \rho \Delta\theta_0. \quad (8)$$

The signal terms $\mu(\lambda r_1 - r^*)$ cancel exactly—both children receive the same signal, and the signal's effect is independent of the level of θ . Iterating forward n periods:

$$D_n = \rho^n \Delta\theta_0. \quad (9)$$

The treatment effect decays geometrically at rate ρ , *regardless* of the post-treatment investment sequence $\{I_1, \dots, I_n\}$. A treated child in an excellent school and a treated child in a terrible school lose the treatment effect at exactly the same rate. This is Formation Invariance.

The derivative that distinguishes the models. The interaction of the treatment effect with post-treatment investment is:

$$\frac{\partial D_n}{\partial I_t} = 0 \quad \forall t \in \{1, \dots, n\} \quad (\text{formation}). \quad (10)$$

Compare with the skills model (from equation 5):

$$\frac{\partial D_n}{\partial I_t} > 0 \quad \text{for some } t \quad (\text{skills, when } \phi < 1). \quad (11)$$

The sign of $\partial D_n / \partial I_t$ is the test. This is the empirical content of the nesting parameter δ introduced in the next section.

2.3 Robustness of Formation Invariance

The zero-interaction prediction of the reduced-form model must survive microfoundational scrutiny. Three derivations follow.

Bayesian learning with a misspecified model. The agent does not know her signal is biased. She updates her belief about the mean return m via a Kalman filter:

$$m_{t+1}^g = (1 - \kappa_t) m_t^g + \kappa_t \tilde{r}_t, \quad \kappa_t = \frac{\sigma_{t|t-1}^2}{\sigma_{t|t-1}^2 + \sigma_\varepsilon^2}. \quad (12)$$

Patience maps linearly to the believed return: $\theta_t = a + b m_t$. The treatment raised the treated child's prior: $m_0^T = m_0^C + \Delta m$. Both children now observe the *same* post-treatment signals $\{\tilde{r}_1, \tilde{r}_2, \dots\}$, with the same noise variance σ_ε^2 . Their posteriors evolve as:

$$\begin{aligned} m_{t+1}^T &= (1 - \kappa_t) m_t^T + \kappa_t \tilde{r}_t, \\ m_{t+1}^C &= (1 - \kappa_t) m_t^C + \kappa_t \tilde{r}_t. \end{aligned}$$

Subtracting: $m_{t+1}^T - m_{t+1}^C = (1 - \kappa_t)(m_t^T - m_t^C)$. The gap in beliefs decays at rate $(1 - \kappa_t)$, which depends on posterior precision σ_t^2 and signal noise σ_ε^2 —both properties of the *agent's learning process*, not of the signal quality $\tilde{r}_t$. The signal quality affects the *level* of both posteriors equally; it does not affect their *difference*. Therefore $\partial(m_t^T - m_t^C) / \partial r_t = 0$, and the preference gap $D_t = b(m_t^T - m_t^C)$ is invariant to the post-treatment environment.

Cultural transmission (Bisin and Verdier, 2001, 2023). A parent of type θ^g invests socialization effort d^g in transmitting her type. With probability d^g the child adopts the parent's type; with probability $1 - d^g$ the child draws from the social distribution with mean $\bar{\theta}_{\text{social}}$. The parent's optimal effort satisfies $d^g = \Delta V^g / c$, where ΔV^g is the perceived value of having a child of one's own type rather than the social default.

For treated and control families in the same community:

- The social default $\bar{\theta}_{\text{social}}$ is the same (both live in the same community).
- The treated parent has $\theta^T = \bar{\theta} + \rho^n \Delta\theta_0$; the control parent has $\theta^C = \bar{\theta}$.
- $\Delta V^T \neq \Delta V^C$ because the treated parent's type differs from the social default by a different amount.

But this differential effort is driven by $D_n = \theta^T - \theta^C$, not by the post-treatment investment I_t . A change in school quality alters $\bar{\theta}_{\text{social}}$ for both families equally, shifting both ΔV^T and ΔV^C by the same amount. The *difference* in socialization effort, to first order, is:

$$d^T - d^C = \frac{\partial \Delta V}{\partial \theta} D_n / c + O(D_n^2). \quad (13)$$

This depends on D_n , not on I_t . The interaction $\partial D_{n+1} / \partial I_t = 0$ at first order. At second order, there is a correction proportional to $D_n \cdot \partial \bar{\theta}_{\text{social}} / \partial I_t$ —the treated parent adjusts effort differently when the social default shifts—but this is $O(D_n^2)$ and small for modest treatment effects.

Chosen preferences (BBMZ). Each agent selects a worldview θ to minimize $(\theta - \theta^*)^2 + \beta(\theta - \theta^{\text{ref}})^2$, where θ^* is the materially optimal patience (a function of the current signal) and $\theta^{\text{ref}} = \rho\theta_t + (1 - \rho)\bar{\theta}$ is the reference (inherited + social baseline). The FOC gives:

$$\theta_{t+1} = \frac{\theta^*(r_t) + \beta \theta^{\text{ref}}}{1 + \beta}. \quad (14)$$

Both treated and control face the same r_t , so $\theta^*(r_t)$ is identical. Their reference points differ: $\theta^{\text{ref},T} - \theta^{\text{ref},C} = \rho D_t$. The gap evolves as:

$$D_{t+1} = \frac{\beta\rho}{1 + \beta} D_t. \quad (15)$$

The effective persistence $\beta\rho / (1 + \beta)$ is a function of identity attachment (β) and cultural persistence (ρ)—neither of which depends on the post-treatment investment I_t . The interaction is zero: $\partial D_{t+1} / \partial I_t = 0$.

Proposition 1 (Formation Invariance). *Under the formation model, and under each of its three microfoundations (Bayesian learning, Bisin-Verdier cultural transmission, BBMZ chosen preferences), the non-cognitive treatment effect is independent of the post-treatment environment to first order:*

$$\frac{\partial D_t}{\partial I_t} = 0 \quad \forall t \geq 1. \quad (16)$$

Remark 1. *The result has a common structure across all three microfoundations: the treatment*

creates a difference between the treated and control agent's internal state (posterior belief, parental type, reference worldview), and the post-treatment environment shifts the level of both agents' states equally, leaving the difference unaffected. The environment is a common shock; the treatment is a differential one. In additive models, common shocks cancel in differences.

In the CES technology, this cancellation fails because the production function is multiplicatively interactive: $f(\theta + \Delta, I) - f(\theta, I) \neq f(\theta + \Delta, I') - f(\theta, I')$ when $\phi < 1$. The treatment effect depends on the level of investment because the CES aggregator couples the stock and the investment through the curvature parameter ϕ . This coupling is precisely what dynamic complementarity means.

3 The Nested Model

3.1 Specification

We embed both models in a single transition function by adding an interaction term to the formation model:

$$\theta_{t+1} = \rho \theta_t + (1 - \rho) \bar{\theta} + \mu(\lambda r_t - r^*) + \delta \theta_t I_t \quad (17)$$

The first three terms are the formation model (equation 6). The last term, $\delta \theta_t I_t$, captures the *interaction* between the current trait and the current investment—the discrete analog of the CES cross-partial (3). The parameter $\delta \geq 0$ is the nesting parameter:

- $\delta = 0$: pure formation. The transition function is additively separable. No complementarity.
- $\delta > 0, \rho > 0$: both mechanisms operate. Persistence comes from cultural transmission (ρ) and from skill-investment complementarity (δI).
- $\delta > 0, \rho = 0$: pure skills. No cultural persistence; the trait persists only because it raises the return to investment.

Connection to the CES. The nested model (17) is not a re-parameterization of the CES. It is a *linearization* of the CES around the baseline stock $\bar{\theta}$ that retains the first-order interaction while adding the additive persistence and adaptation terms that the CES lacks. Specifically, a first-order Taylor expansion of $f(\theta_t, I_t) = [\gamma \theta_t^\phi + (1 - \gamma) I_t^\phi]^{1/\phi}$ around $(\bar{\theta}, \bar{I})$ gives:

$$f(\theta_t, I_t) \approx f(\bar{\theta}, \bar{I}) + f_\theta \cdot (\theta_t - \bar{\theta}) + f_I \cdot (I_t - \bar{I}) + f_{\theta I} \cdot (\theta_t - \bar{\theta})(I_t - \bar{I}). \quad (18)$$

The self-productivity term $f_\theta(\theta_t - \bar{\theta})$ maps to $\rho(\theta_t - \bar{\theta})$; the investment response $f_I(I_t - \bar{I})$ maps to $\mu(\lambda r_t - r^*)$ (since the return signal is proportional to the investment environment); and the cross-partial $f_{\theta I}(\theta_t - \bar{\theta})(I_t - \bar{I})$ maps to $\delta\theta_t I_t$ (absorbing the centering into the intercept). The nested model thus captures the leading-order content of the CES technology while adding the mean-reversion and signal-adaptation terms that the CES omits.

3.2 Treatment Effects and Identification

Under the nested model, the treatment effect evolves as:

$$D_{t+1} = \theta_{t+1}^T - \theta_{t+1}^C = (\rho + \delta I_t) D_t. \quad (19)$$

The *effective persistence* is $\rho_{\text{eff}}(I_t) = \rho + \delta I_t$: a sum of the cultural persistence rate and the investment-dependent complementarity rate. Iterating forward n periods with investment sequence $\{I_1, \dots, I_n\}$:

$$D_n = D_0 \prod_{t=1}^n (\rho + \delta I_t). \quad (20)$$

Taking logs:

$$\ln D_n = \ln D_0 + \sum_{t=1}^n \ln(\rho + \delta I_t). \quad (21)$$

For δ small relative to ρ (which is the null of formation dominance):

$$\ln D_n \approx \ln D_0 + n \ln \rho + \frac{\delta}{\rho} \sum_{t=1}^n I_t + O(\delta^2). \quad (22)$$

The log treatment effect is approximately linear in the cumulative post-treatment investment. The *intercept* ($n \ln \rho + \ln D_0$) identifies ρ from the average treatment effect in the low-investment subsample. The *slope* (δ/ρ) identifies the complementarity parameter from the gradient of the treatment effect with respect to cumulative investment quality.

Proposition 2 (Separate Identification). *In the nested model (17), ρ and δ are separately identified when there is cross-sectional variation in post-treatment investment quality I_t :*

- (a) ρ is identified from the treatment effect among children in the lowest-quality post-treatment environments: as $I_t \rightarrow 0$, effective persistence converges to ρ .
- (b) δ is identified from the treatment $\times$ post-treatment-quality interaction: $\partial \ln D_n / \partial \bar{I} = n\delta/\rho$.

(c) The test $H_0 : \delta = 0$ (formation) versus $H_1 : \delta > 0$ (skills) is equivalent to the test of a zero coefficient on $T_i \times Q_i$ in the reduced-form regression:

$$y_i = \alpha + \beta_1 T_i + \beta_2 Q_i + \beta_3 (T_i \times Q_i) + X_i \gamma + \varepsilon_i, \quad (23)$$

where T_i is random treatment assignment, Q_i is post-treatment environmental quality, and β_3 is the reduced-form analog of δ . Under formation: $\beta_3 = 0$. Under skills: $\beta_3 > 0$.

Remark 2. The key identifying assumption is that post-treatment environmental quality Q_i is independent of the treatment effect $\Delta\theta_0$, conditional on covariates. In the HSIS, treatment assignment is random, so $T_i \perp Q_i | X_i$ holds by design. The threat is that treated children sort into different schools (higher θ leads to better school choices), which would generate $\beta_3 > 0$ even under formation. We address this in Section 7 with an IV approach using district-level school quality variation.

3.3 The Dual-Process Extension

The nested model can be extended to allow different δ parameters for cognitive and noncognitive dimensions:

$$\theta_{t+1}^C = \rho^C \theta_t^C + (1 - \rho^C) \bar{\theta}^C + \mu^C I_t^C + \delta^C \theta_t^C I_t^C, \quad (24)$$

$$\theta_{t+1}^N = \rho^N \theta_t^N + (1 - \rho^N) \bar{\theta}^N + \mu^N \lambda r_t + \delta^N \theta_t^N I_t^N. \quad (25)$$

The cognitive equation (24) treats instruction (I_t^C) as the driving input, with μ^C governing the direct response. The noncognitive equation (25) treats the return signal (λr_t) as the driving input, with μ^N governing adaptation. The key notational difference reflects the dual-process hypothesis: cognitive outcomes respond to *instructional investment* (a flow that can be turned on and off), while noncognitive outcomes respond to the *signal environment* (the park, which deposits a preference).

The dual-process hypothesis is:

$$\delta^C > 0, \quad \delta^N = 0. \quad (26)$$

Cognitive outcomes exhibit complementarity with post-treatment investment; noncognitive outcomes do not. This is a sharper claim than either “skills dominate” ($\delta^C > 0, \delta^N > 0$) or “formation dominates” ($\delta^C = 0, \delta^N = 0$). It says both are right—about different things.

3.4 Decomposing Persistence: The Formation Share

When both mechanisms operate (Outcome C: $\rho > 0$ and $\delta > 0$), the policy question is not which mechanism exists but which *dominates quantitatively*. The nested model provides a natural decomposition.

From equation (19), the effective persistence is $\rho_{\text{eff}} = \rho + \delta \bar{I}$, where $\bar{I}$ is the average post-treatment investment. Define the *formation share*:

$$\omega_F = \frac{\rho}{\rho + \delta \bar{I}}, \quad \omega_S = \frac{\delta \bar{I}}{\rho + \delta \bar{I}} = 1 - \omega_F. \quad (27)$$

The formation share ω_F is the fraction of noncognitive persistence attributable to cultural transmission, and ω_S is the fraction attributable to dynamic complementarity. Both are identified once $\hat{\rho}$ and $\hat{\delta}$ are separately estimated (Proposition 2).

The decomposition converts the binary test ($\delta = 0$ vs. $\delta > 0$) into a continuous measure of relative importance. Even if $\hat{\delta}^N > 0$ at some margin, the policy conclusion depends on ω_F :

- $\omega_F > 0.8$: persistence is dominated by cultural transmission. Concentrate resources on the signal environment during the formation window.
- $\omega_F \approx 0.5$: both mechanisms contribute equally. Invest in both the formation window and sustained post-program quality.
- $\omega_F < 0.2$: persistence is dominated by complementarity. Spread investment over the full K–12 horizon.

From equation (22), the formation share can be estimated directly as:

$$\hat{\omega}_F = 1 - \frac{\hat{\delta}/\hat{\rho} \cdot \bar{I}}{1 + \hat{\delta}/\hat{\rho} \cdot \bar{I}} = \frac{1}{1 + (\hat{\delta}/\hat{\rho})\bar{I}}. \quad (28)$$

This requires only the ratio $\hat{\delta}/\hat{\rho}$ (identified from the slope of the log treatment effect with respect to cumulative investment, equation 22) and the mean investment $\bar{I}$ (observed). Standard errors on $\hat{\omega}_F$ follow by the delta method.

The decomposition also determines the *optimal investment horizon*. Under pure formation ($\omega_F = 1$), a dollar invested in year 1 has the same long-run effect as a dollar invested in year 5 of the child's life—both shift the signal during the formation window, and neither interacts with subsequent investment. The optimal policy concentrates spending in the formation window (ages 0–5) where the Kalman gain is highest. Under pure skills ($\omega_F = 0$), a dollar in year 1 has a *higher* long-run effect than a dollar in year 5 because it compounds through complementarity with 17 years of subsequent schooling rather than

13. The optimal policy front-loads spending but sustains it throughout childhood.

At intermediate ω_F , the optimal allocation trades off signal intensity (favoring concentration) against complementarity (favoring sustained investment). Section 8 derives this tradeoff explicitly.

3.5 How the Program Produces $\Delta\theta_0$: Cross-Program Predictions

The nested model takes $\Delta\theta_0$ as given. But the four landmark experiments—Perry, Abecedarian, Jamaica, Head Start—differ dramatically in design, intensity, and cost. The formation model generates specific predictions about which program features drive $\Delta\theta_0$ and which do not.

Under formation, the initial deposit is determined by the *signal contrast* between the treatment and control environments, the *duration* of exposure, and the *adaptation speed* during the formation window:

$$\Delta\theta_0 = \mu \cdot A \cdot (\lambda_{\text{treat}} - \lambda_{\text{control}}) r^*, \quad (29)$$

where A is the number of formative periods of exposure and $\lambda_{\text{treat}} - \lambda_{\text{control}}$ is the signal contrast—how much more information the treated child receives about the returns to patience and investment relative to the control child. The signal contrast depends on who the child interacts with (role models), what environments she is exposed to (peer group, curriculum, physical setting), and how different these are from what she would have experienced without the program.

Under skills, the initial deposit is determined by the *total investment volume*:

$$\Delta\theta_0 = h\left(\sum_{t=1}^A I_t\right), \quad (30)$$

where h is the CES production function and I_t is the investment per period. More hours, more years, more resources, more trained teachers all raise $\Delta\theta_0$ through the production function.

The two models make different predictions about the *production function for initial effects*: formation says signal contrast per unit time drives $\Delta\theta_0$; skills says investment volume drives $\Delta\theta_0$. This generates a cross-program test.

Consider the four programs:

Program	Duration	Hours/week	Cost/child	Long-run effect
Perry	2 years	12.5	~\$20K	B/C 7–10
Abecedarian	5 years	40+	~\$90K	B/C 7–13
Jamaica	2 years	1	~\$1.5K	Wages +43%
Head Start	1–2 years	17.5	~\$8K	Mixed

Jamaica is the decisive case. One hour per week for two years—a community health worker visiting the home, playing with the child, modeling enriched parent-child interaction—produced adult earnings effects comparable to Perry’s two years of half-day instruction and Abecedarian’s five years of full-day care. Under the skills model, Jamaica invested approximately $1/60$ the resources of Abecedarian per child. The production function $h(\cdot)$ would need extreme concavity (the first hour is $60\times$ more productive than the 41st hour in the same week) to rationalize comparable effects from such different inputs. Under the formation model, the explanation is simpler: Jamaica had *extremely high signal contrast per unit time*. A trained visitor entering a home where stimulation was near-zero—where stunted children spent their days in unstimulating environments—provided a signal about the returns to parent-child engagement that was radically different from the child’s baseline experience. One hour per week of this maximally contrastive signal was enough to shift $\lambda_{\text{treat}} - \lambda_{\text{control}}$ by a large amount, despite the low total hours.

Prediction 1 (Cross-Program). *Under formation: effect size per dollar is determined by signal contrast, not by investment volume. Programs with high contrast and low cost (Jamaica) should produce returns per dollar comparable to programs with moderate contrast and high cost (Abecedarian). Under skills: effect size per dollar is determined by the production function’s returns to scale. Programs with more investment hours should dominate.*

The published benefit-cost ratios are consistent with formation. García et al. (2020) estimate Abecedarian’s IRR at 13.7% per annum (\$90K investment). Gertler et al. (2021) estimate Jamaica’s wage return at 43% on a \$1.5K investment. The Jamaica return per dollar invested exceeds Abecedarian’s by roughly an order of magnitude. This is difficult to rationalize with a concave production function of investment volume; it is natural if what matters is the signal contrast and the formation window was open in both programs.

A further prediction: within Head Start, the variation in effect sizes across centers should correlate with the *contrast* between the Head Start experience and the counterfactual (the care the child would have received otherwise), not with the total hours of instruction. García et al. (2020) note that 75% of Abecedarian control children attended alternative childcare—so the “treatment” is the difference between Abecedarian care and

alternative care, not the difference between Abecedarian and no care. Centers where the control condition is poor (high contrast) should show larger effects than centers where the control condition is rich (low contrast), holding program quality fixed. Kline and Walters (2016) find exactly this in the HSIS: the treatment effect is largest when Head Start substitutes for no care and smallest when it substitutes for another preschool program. This is the formation prediction: what matters is the signal gap, not the investment level.

4 Five Discriminating Predictions

Each prediction is derived from the structural differences between the models and mapped to a testable moment.

4.1 Prediction 1: The Complementarity Test

This follows directly from Proposition 2(c).

Prediction 2 (Complementarity). *Estimate β_3 in (23) separately for cognitive and noncognitive outcomes:*

- *Pure skills: $\beta_3^C > 0$ and $\beta_3^N > 0$. Both outcome types interact with post-treatment quality.*
- *Pure formation: $\beta_3^C = 0$ and $\beta_3^N = 0$. Neither outcome type interacts.*
- *Dual process: $\beta_3^C > 0$ and $\beta_3^N = 0$. Cognitive outcomes interact; noncognitive outcomes do not.*

The dual-process prediction (26) is uniquely generated by the model in which cognitive outcomes follow the skills technology and noncognitive outcomes follow the formation mechanism.

4.2 Prediction 2: Wealth vs. Earnings

Under the skills model, the noncognitive treatment raises productivity A , which raises wages $w = A \cdot g(\text{human capital})$. The agent saves at a fixed rate σ , so wealth grows proportionally to earnings: $W = \sigma \sum_t w_t / (1 + r)^t$. The wealth-to-earnings ratio W/Y is determined by σ and r , neither of which changes with treatment:

$$\frac{W_T/Y_T}{W_C/Y_C} = \frac{\sigma_T}{\sigma_C} = 1 \quad (\text{skills: preferences are fixed}). \quad (31)$$

Under the formation model, the treatment raised θ , and θ governs the saving rate $\sigma(\theta) = \theta / (1 + \theta)$. Higher θ produces both higher earnings (through better occupational

choice) and a higher saving rate (through greater patience). Wealth accumulates disproportionately:

$$\frac{W_T/Y_T}{W_C/Y_C} = \frac{\sigma(\theta_T)}{\sigma(\theta_C)} = \frac{\theta_T(1 + \theta_C)}{\theta_C(1 + \theta_T)} > 1 \quad (\text{formation}). \quad (32)$$

Prediction 3 (Wealth Test). Define the wealth multiplier $\Delta = (W_T/Y_T)/(W_C/Y_C)$. Under skills: $\Delta = 1$. Under formation: $\Delta > 1$. The excess $\Delta - 1$ identifies the treatment effect on the saving rate, which exists only if the treatment changed preferences (not just productivity).

4.3 Prediction 3: Between vs. Within Occupation

Under the skills model, higher noncognitive skills raise productivity *within* any occupation. Conditional on the same occupation, treated workers should earn more than control workers because their skills make them more effective. The treatment effect should have both a *between*-occupation component (treated workers sort into higher-paying jobs) and a *within*-occupation component (treated workers earn more in any given job).

Under the formation model, θ governs *choices*, not productivity. A more patient agent chooses a higher-return occupation, invests more in education, and avoids high-present-reward/high-future-cost activities (crime). But conditional on the same choice, she is no more productive than a control agent with the same choice. The treatment effect operates entirely through the between-occupation channel.

Formally, decompose earnings as $Y_i = \sum_j \mathbf{1}[j_i = j] w_j + v_i$, where j_i is occupation and v_i is the within-occupation residual. The treatment effect decomposes as:

$$\mathbb{E}[Y_T - Y_C] = \underbrace{\sum_j w_j (P_T(j) - P_C(j))}_{\text{between: occupational sorting}} + \underbrace{\sum_j P_T(j) \mathbb{E}[v_T|j] - \sum_j P_C(j) \mathbb{E}[v_C|j]}_{\text{within: conditional productivity}}. \quad (33)$$

Prediction 4 (Mediation). Under skills: the within-occupation component is positive ($\mathbb{E}[v_T|j] > \mathbb{E}[v_C|j]$). Under formation: the within-occupation component is zero ($\mathbb{E}[v_T|j] = \mathbb{E}[v_C|j]$). The test is whether the treatment effect survives conditioning on occupation.

4.4 Prediction 4: Cognitive vs. Non-Cognitive Trajectories

The skills model treats both cognitive and noncognitive outcomes as stocks that depreciate:

$$\Delta C_a = \Delta C_0 \cdot \prod_{t=1}^a \rho_{\text{eff}}^C(I_t^C), \quad (34)$$

$$\Delta N_a = \Delta N_0 \cdot \prod_{t=1}^a \rho_{\text{eff}}^N(I_t^N). \quad (35)$$

Both follow the same process (equation 20); they differ only in parameters. The trajectories should be qualitatively similar: smooth decay, with the rate governed by the effective persistence at each stage.

The formation model treats them as different processes:

$$\text{Cognitive: } C_a = \alpha + \mu^C \cdot I_a^C + \varepsilon_a. \quad (36)$$

Cognitive performance responds to the *current* instructional environment—a flow. During the program ($a \leq a^*$), I_a^C is high and C_a is elevated. After the program, I_a^C drops to whatever the regular school provides, and C_a drops with it. The “fadeout” is not depreciation of a stock; it is removal of an instructional input.

$$\text{Noncognitive: } N_a = \rho^{a/g} \cdot \Delta\theta_0 + \mu^N(\lambda r_a - r^*). \quad (37)$$

The noncognitive outcome reflects two components: the decaying deposit ($\rho^{a/g}\Delta\theta_0$, where g is the generational period) and the current signal. The deposit is the formation channel; the current signal is the enforcement channel. At short horizons, both contribute. At long horizons, the deposit dominates if ρ is high.

Prediction 5 (Two Processes). *Under skills: cognitive and noncognitive trajectories are qualitatively similar (both stocks, both decaying, different rates). Under formation: they are qualitatively different. Cognitive outcomes show a step function—elevated during the program, dropping when the program ends, sustained only if the subsequent school maintains the instructional input. Noncognitive outcomes show smooth geometric decay from the treatment endpoint, independent of the subsequent environment. The signature: cognitive fadeout interacts with post-treatment school quality; noncognitive persistence does not.*

Remark 3. *This prediction connects Predictions 1 and 4: the complementarity test (Prediction 1, $\beta_3^N = 0$) and the trajectory test (Prediction 4, cognitive step-down vs. noncognitive smooth decay) are two manifestations of the same structural distinction. If the dual-process hypothesis (26)*

is correct, both should hold simultaneously: cognitive outcomes interact with post-treatment quality and show step-function fadeout; noncognitive outcomes do not interact and show geometric persistence. Observing one without the other would challenge the dual-process interpretation.

4.5 Prediction 5: Gender Differential

Under the skills model, a gender gap in treatment effects reflects gender-differential *returns* to noncognitive skills in the labor market. If women’s wages are more sensitive to conscientiousness, patience, or self-control than men’s wages, the same skill increment $\Delta\theta$ produces a larger earnings gain for women. The gender gap should *survive* conditioning on noncognitive traits: $\mathbb{E}[Y^F|\theta, T] - \mathbb{E}[Y^M|\theta, T] > 0$ even at the same θ .

Under the formation model, a gender gap reflects gender-differential $\Delta\theta$. If the program changed the signal environment more for girls—for instance, because the teachers modeled a female career path that was absent from the home environment—then $\Delta\theta^F > \Delta\theta^M$. But conditional on the same $\Delta\theta$, the earnings response is identical across genders: $\mathbb{E}[Y|\theta, T, \text{female}] = \mathbb{E}[Y|\theta, T, \text{male}]$.

Prediction 6 (Gender). *Regress earnings on treatment, non-cognitive mediators, and their gender interactions. Under skills: a significant gender $\times$ treatment interaction survives after conditioning on non-cognitive traits. Under formation: the gender interaction shrinks toward zero after conditioning, because the gender gap in outcomes is mediated entirely by the gender gap in $\Delta\theta$.*

5 Evidence: Published Tables and Figures

This section maps each prediction to specific published results. I reference tables and figures by their identifiers in the original publications.

5.1 The Complementarity Test: Mixed but Informative

The existing quasi-experimental evidence on Prediction 2 is genuinely contested. Three sets of results bear directly on the test; they disagree.

Evidence for $\delta > 0$: Johnson and Jackson (2019). Johnson and Jackson (2019) exploit county-level variation in Head Start rollout (1965–1980) and court-ordered school finance reforms (1970s–1990s) using PSID data. Their central finding (Table 5, p. 335): Head

Start exposure raises educational attainment by 0.22 years and wages by 5.6% in 75th-percentile K–12 spending districts, but only 0.10 years and 1.9% in 25th-percentile districts. The interaction is statistically significant and economically large—the return to Head Start roughly doubles with high subsequent school spending. This is evidence for $\delta > 0$ and is the strongest published support for dynamic complementarity.

However, the Johnson-Jackson design identifies the interaction between *aggregate* early-childhood exposure and *aggregate* school spending at the county level, not the interaction between an individual treatment and individual-level school quality. The complementarity could operate through community channels (counties with both good preschool and good schools are systematically different) rather than through individual-level skill-investment interaction. The formation model can accommodate their finding if the county-level interaction reflects the *signal environment* rather than the investment technology: counties with both Head Start and high school spending provide a richer developmental signal, depositing higher θ through the $\mu(\lambda r - r^*)$ channel. The interaction would then be in the formation equation’s signal component, not in the complementarity parameter δ .

Evidence for $\delta \leq 0$: Goff et al. (2025) and Rossin-Slater and Wüst (2020). Goff et al. (2025) use Romanian data combining a regression discontinuity in school quality with a differences-in-differences design exploiting the 1966 abortion ban as a shock to family environments. They find “no consistent evidence of interactions” between family environment quality and school quality (Table 5), with point estimates that are “suggestive of substitutability.” Rossin-Slater and Wüst (2020) study Danish data and find that prior exposure to a nurse home visiting program *reduces* the added value of preschool by 88% for educational attainment and 80% for survival past 65—programs function as partial substitutes. These findings are inconsistent with $\delta > 0$ and consistent with either $\delta = 0$ (formation) or $\delta < 0$ (substitutability, which neither model predicts in its baseline form).

Direct RCT evidence: Schiman (2024). Schiman (2024), using HSIS microdata (the only large-scale early childhood RCT in the US), directly tests whether preschool skill gains interact with subsequent school-year investment. Her finding: “effects of preschool depreciated significantly within a year and there were no positive dynamic complementarities that offset depreciation.” This is the closest existing test to the specification in (23), and it finds $\hat{\delta} \approx 0$ —the formation prediction.

Assessment. The quasi-experimental evidence is split, but the *RCT* evidence—from the only study with both random treatment assignment and detailed post-treatment environment data—favors $\delta = 0$. The Johnson-Jackson finding may reflect community-level signals rather than individual-level complementarity. The key unresolved question is whether their interaction operates on cognitive or non-cognitive margins. If it operates on cognitive outcomes (where the skills model applies), it is consistent with the dual-process interpretation: $\delta > 0$ for cognitive skills, $\delta = 0$ for preferences. Neither Johnson-Jackson nor Schiman decompose by outcome type—this is the empirical contribution that remains to be made.

5.2 Wealth vs. Earnings: Perry at Age 40

Schweinhart et al. (2005) report age-40 outcomes separately by treatment arm and gender (Summary Table, pp. xv–xvi). For males:

Outcome	Treatment	Control	Ratio
Median monthly earnings	\$2,020	\$1,460	1.38
Home ownership	37%	28%	1.32
Savings account	76%	50%	1.52
Second car ownership	30%	13%	2.31

The earnings ratio is 1.38. Asset accumulation ratios range from 1.32 (home ownership) to 2.31 (second car). The asset-to-earnings multiplier Δ is 1.0–1.7, depending on the asset measure. Under the skills model with fixed saving rates, all ratios should be approximately 1.38 (proportional to earnings). The excess in durable goods accumulation—particularly the $2.31 \times$ ratio for second cars, a discretionary saving/consumption choice—is consistent with the formation prediction: treated males have a higher saving rate $\sigma(\theta_T) > \sigma(\theta_C)$, producing disproportionate asset accumulation.

The female outcomes show an even sharper pattern: high school graduation rates of 84% vs. 32% (treatment vs. control), a $2.6 \times$ ratio, while the earnings differential is smaller. The education gap exceeds the earnings gap—consistent with $\Delta\theta$ changing the *decision* to invest in schooling, not just the productivity within a given education level.

5.3 Occupational Mediation: Heckman, Pinto, and Savelyev (2013)

Heckman et al. (2013) decompose Perry’s adult outcomes through three latent factors measured at ages 7–9: cognition, externalizing behavior, and academic motivation. Their central finding (Figures 6–7, pp. 2075–2076; Table 6, p. 2074):

For males, the treatment had *no significant effect on cognition* ($p = 0.683$) but significantly reduced externalizing behavior ($p = 0.038$). Externalizing behavior accounted for 20–60% of the treatment effects on crime, employment, and income. The channel is behavioral sorting: treated males made different choices (less crime, more employment), not different within-occupation productivity.

For females, both externalizing behavior and academic motivation contributed: externalizing behavior mediated 40–60% of crime reductions; academic motivation mediated 30–35% of cognitive and mental health outcomes. The treatment operated through multiple non-cognitive channels, all of which affect *choices* (occupational selection, educational investment, criminal behavior) rather than within-choice productivity.

The skills model predicts a significant *direct* effect of treatment on earnings after conditioning on occupational choice: if non-cognitive skills raise within-occupation productivity, the direct channel should survive mediation. Heckman, Pinto, and Savelyev’s decomposition shows that the direct channel is small and often insignificant—the treatment effect operates *through* changed behavior, not through changed productivity within a behavior. This is the formation prediction.

5.4 Cognitive vs. Non-Cognitive Trajectories: The Dual-Process Signature

This is the most revealing evidence, and it requires careful attention to recent findings that complicate the standard narrative.

Perry cognitive trajectory. Schweinhart et al. (2005) report Stanford-Binet IQ treatment effects at multiple ages: +11 points at age 5 (during program), declining to +6 at age 6, +4 at age 8, and statistically insignificant by age 10. By age 14 (WISC), treatment and control scored 80 and 81 respectively. The cognitive gain disappears within 5 years of program end.

Perry non-cognitive trajectory. Behavioral treatment effects (teacher-reported) are significant at ages 6–9. Employment, savings, and crime outcomes are significant at ages 19, 27, and 40—approximately stable magnitudes relative to the control mean. The non-cognitive channel persists for 35+ years after a 2-year intervention.

Complication 1: Perry at age 54. García et al. (2023) report that at age 54, Perry treatment effects on *cognition* are significant: executive functioning +0.36 SD (males: +0.56

SD), openness +0.32 SD, positive personality +0.42 SD. This contradicts simple permanent cognitive fadeout. However, it is consistent with the formation model if the non-cognitive deposit (θ) eventually *produces* cognitive gains through better life choices: higher $\theta \rightarrow$ more education $\rightarrow$ more cognitively stimulating occupations $\rightarrow$ higher age-54 cognition. The cognitive gains at 54 are downstream of the non-cognitive deposit, not a recovery of the original cognitive stock. The formation model predicts this: the park deposited θ (noncognitive), θ changed choices, choices changed the lifetime cognitive environment. The cognitive emergence at 54 is *mediated* by the behavioral channel, not a reassertion of the original cognitive investment.

Complication 2: Jamaica’s cognitive reemergence. The Jamaica Early Childhood Stimulation study shows an even more striking pattern. IQ treatment effects were significant during the intervention (ages 9–24 months through age 2), *not significant at age 7* (Gertler et al., 2014), but *reappeared* at ages 11 and 17 and were sustained through age 31 (Hedges $g = 0.57$ at age 31; Walker et al. 2022). Treatment effects on labor market outcomes *grew* from 25% higher earnings at age 22 to 43% higher wages at age 31 (Gertler et al., 2021). This “growing treatment effect” is inconsistent with simple exponential decay from either model but consistent with the formation model’s cascade mechanism: the intervention deposited θ , θ led to better choices, better choices led to a richer environment, which eventually produced cognitive gains through sustained engagement—a delayed positive feedback from the noncognitive deposit.

Complication 3: Hart et al. (2024). A recent meta-analysis of 86 RCTs (56,662 participants) by Hart et al. (2024) finds that *both* social-emotional and cognitive impacts demonstrate fadeout, and at 1–2 year follow-up, “cognitive skills showed greater conditional persistence than social-emotional skills”—contrary to the standard Heckman narrative. This is a serious challenge to any model that assumes noncognitive gains selectively persist at short-to-medium horizons. However, the meta-analysis covers 1–2 year follow-up windows, not the 35–40 year horizons where Perry’s non-cognitive effects are observed. The formation model predicts that *within-lifetime* fadeout of observable non-cognitive measures is consistent with preference persistence: the behavioral expressions of θ (which surveys measure) may attenuate as the child adapts to new environments, while the underlying θ deposit decays slowly at rate ρ . The deposit is not the measured behavior; it is the latent preference that produces the behavior in interaction with the environment. At short horizons, environment shifts dominate and measures fade; at long horizons, the persistent θ reasserts itself as accumulated behavioral choices compound.

Assessment. The trajectory evidence reveals a more nuanced picture than either model in its simple form predicts. Cognitive outcomes show fadeout-then-reemergence (Perry at 54, Jamaica at 11+)—inconsistent with simple exponential decay and consistent with a noncognitive deposit that eventually produces cognitive gains through changed choices. Non-cognitive outcomes show short-run fadeout (Hart et al. 2024) but long-run persistence (Perry at 40, Jamaica at 31)—consistent with a preference deposit that is not fully captured by behavioral surveys at short horizons but manifests through accumulated life decisions at long horizons. The formation model accommodates both patterns; the skills model requires additional mechanisms (reemergence through cross-productivity, different fadeout horizons for different skill dimensions) that stretch the framework beyond its original parsimony.

5.5 Gender Differential: Differential $\Delta\theta$ vs. Differential Returns

Heckman et al. (2013) report treatment effects separately by gender. The education differential is dramatic: female high school graduation treatment effect is +52 percentage points (84% vs. 32%), while the male effect is modest and insignificant. Crime effects are male-concentrated: 5+ lifetime arrests drop from 55% to 36% for treated males. Their mediation analysis (Table 6, p. 2074) shows that conditioning on the three non-cognitive factors substantially reduces the gender gap in treatment effects.

The Perry teachers were Black women with professional careers—college-educated and employed in the 1960s, a demographic rarity in the Ypsilanti community. For girls, these teachers modeled a female future absent from the home environment; for boys, they modeled female professional success, which is informative but not gender-matched. The formation model predicts that the park changed θ more for girls (larger signal contrast on the gender-relevant dimension). The skills model predicts that girls have higher returns to non-cognitive skills in the labor market (the same $\Delta\theta$ produces larger earnings gains for women).

The Heckman-Pinto-Savelyev decomposition is consistent with the formation prediction: the gender differential is mediated by different magnitudes of non-cognitive change, not by different returns to the same change. Conditional on the same non-cognitive factor scores, the gender gap in outcomes narrows substantially.

Cross-program evidence provides a further check. The Jamaica intervention was delivered by community health workers—a more gender-balanced delivery. Gertler et al. (2021) find more gender-balanced treatment effects than Perry. If the formation mechanism (gender-matched role models produce larger $\Delta\theta$) is correct, gender balance in deliv-

ery should produce gender balance in effects. It does.

5.6 Summary of Evidence

Table 1: Five Predictions and Published Evidence

Test	Skills	Formation	Published Evidence
Complementarity	$\beta_3 > 0$	$\beta_3 = 0$	Split: Johnson-Jackson $\beta_3 > 0$; Schiman, Goff et al. $\beta_3 \leq 0$
Wealth/earnings	$\Delta \approx 1$	$\Delta > 1$	Perry male asset ratios $1.3\text{--}2.3\times$ vs. earnings $1.4\times$ (Schweinhart 2005, Summary Table)
Mediation	Direct > 0	Direct ≈ 0	Direct ≈ 0 (HPS 2013, Table 6, Figures 6–7)
Trajectories	Parallel decay	Dual process	Cog fadeout + reemergence; noncog short-run fade + long-run persist (Schweinhart 2005; García et al. 2023; Hart et al. 2024)
Gender gap	Survives	Shrinks	Shrinks after mediation (HPS 2013, Table 6); cross-program: gender-balanced delivery $\rightarrow$ gender-balanced effects (Jamaica vs. Perry)

The complementarity test is genuinely split. The remaining four tests favor formation. The joint pattern—especially the dual-process trajectory evidence and the mediation decomposition—is difficult to reconcile with a pure skills model but natural under the formation interpretation.

6 Estimation Strategy

This section specifies the estimation in detail sufficient for replication. It also addresses, in sequence, six objections that a defender of the skill formation framework would raise against the identification strategy. The objections are real—several would, if left unanswered, invalidate the test. The answers sharpen the design.

6.1 The Primary Test: HSIS with Outcome-Type Decomposition

The Head Start Impact Study randomized approximately 5,000 children across 351 centers in 22 states, with follow-up through 3rd grade (Puma et al., 2012). The microdata are available through ICPSR (Study 36968) under restricted-use agreement. The data contain: random assignment to Head Start (T_i); cognitive outcomes (PPVT, Woodcock-Johnson Letter-Word and Applied Problems) and noncognitive outcomes (teacher-rated Aggressive Behavior, Hyperactive Behavior, Social Skills and Competence, Positive Approaches to Learning) at end-of-Head-Start, 1st grade, and 3rd grade; center identifiers that can be linked to school-quality measures via the NCES Common Core of Data (school-level expenditure per pupil, student-teacher ratios, Title I receipt, value-added estimates where available).

The specification is:

$$y_{it}^k = \alpha^k + \beta_1^k T_i + \beta_2^k Q_{i,\text{post}} + \beta_3^k (T_i \times Q_{i,\text{post}}) + X_i \gamma^k + \varepsilon_{it}^k, \quad (38)$$

estimated separately for cognitive outcomes ($k = C$) and noncognitive outcomes ($k = N$), at each follow-up wave ($t \in \{\text{end-of-HS}, \text{1st grade}, \text{3rd grade}\}$). $Q_{i,\text{post}}$ is the quality of the elementary school the child entered after Head Start, measured by school-level expenditure per pupil or student-teacher ratio. X_i includes baseline demographics, pre-treatment cognitive and noncognitive scores, center fixed effects, and cohort indicators.

The test: $H_0^N : \beta_3^N = 0$ (formation) versus $H_1^N : \beta_3^N > 0$ (skills) for noncognitive outcomes. Separately: $H_0^C : \beta_3^C = 0$ versus $H_1^C : \beta_3^C > 0$ for cognitive outcomes.

The dual-process hypothesis (26) predicts: reject H_0^C (cognitive outcomes interact with school quality) but fail to reject H_0^N (noncognitive outcomes do not). Schiman (2024) estimated a version of this specification without the outcome-type decomposition and found no aggregate complementarity. The contribution here is the decomposition: the aggregate null may mask a positive cognitive interaction offset by a zero noncognitive interaction.

6.2 Objection 1: “You Are Testing a Linearization, Not My Model”

The nested model (17) approximates the CES with a bilinear interaction term $\delta \theta_t I_t$. The CES cross-partial (3) is a nonlinear function of θ_t , I_t , and ϕ —the complementarity is strongest when the stock and investment are imbalanced and weakest when they are proportional. A zero $\hat{\delta}$ might reject the linear approximation without rejecting the nonlinear CES.

Response. The paper includes a direct CES robustness check. Using the HSIS microdata, estimate the CES technology (1) separately for cognitive and noncognitive outcomes via nonlinear least squares:

$$y_{i,t+1}^k = \left[\hat{\gamma}^k (y_{it}^k)^{\hat{\phi}^k} + (1 - \hat{\gamma}^k) (Q_{i,\text{post}})^{\hat{\phi}^k} \right]^{1/\hat{\phi}^k} + \varepsilon_{it}^k, \quad (39)$$

where y_{it}^k is the outcome at the end of Head Start (the “stock”) and $Q_{i,\text{post}}$ is the post-treatment school quality (the “investment”). The test is $H_0 : \hat{\phi}^N = 0$ (Cobb-Douglas, no complementarity for noncognitive outcomes). If $\hat{\phi}^N$ is not significantly different from zero—or is positive (substitutability)—the CES confirms the linearized result.

Additionally, following Agostinelli and Wiswall (2025), we estimate the translog specification:

$$\ln y_{i,t+1}^k = \alpha_0 + \alpha_1 \ln y_{it}^k + \alpha_2 \ln Q_i + \alpha_3 (\ln y_{it}^k)(\ln Q_i) + \alpha_4 (\ln y_{it}^k)^2 + \alpha_5 (\ln Q_i)^2 + \varepsilon_i, \quad (40)$$

which nests Cobb-Douglas ($\alpha_3 = \alpha_4 = \alpha_5 = 0$) without restricting the elasticity of substitution to be constant. The coefficient α_3 is the translog analog of δ : the interaction between log-stock and log-investment. Under formation: $\alpha_3^N = 0$. Under skills: $\alpha_3^N \neq 0$.

This addresses the linearization critique: the test is conducted in the linear, CES, and translog specifications. If all three find $\hat{\delta}^N = 0$, $\hat{\phi}^N \geq 0$, and $\hat{\alpha}_3^N = 0$, the result is not an artifact of functional form.

6.3 Objection 2: “School Quality Is Not Total Investment”

In Cunha et al. (2010), I_t is a latent factor extracted from multiple indicators: parental cognitive stimulation, emotional support, financial expenditure, and school quality. School quality is one component. Complementarity between preschool skills and *parental* investment—which is unobserved in the HSIS—could exist even if the school-quality interaction is zero.

Response. The HSIS data contain home environment measures from the Family and Child Experiences Survey (FACES): the HOME inventory short form, parenting warmth and discipline scales, and the frequency of literacy activities at home. These are imperfect measures of parental investment, but they provide additional variation beyond school quality. The enriched specification is:

$$y_{it}^k = \alpha^k + \beta_1^k T_i + \beta_2^k Q_i^S + \beta_3^k Q_i^H + \beta_4^k (T_i \times Q_i^S) + \beta_5^k (T_i \times Q_i^H) + X_i \gamma^k + \varepsilon_{it}^k, \quad (41)$$

where Q^S is school quality and Q^H is home environment quality. The formation model predicts $\beta_4^N = \beta_5^N = 0$: the noncognitive treatment effect does not interact with *either* school or home investment. The skills model predicts $\beta_4^N > 0$ or $\beta_5^N > 0$ or both. This doubles the power of the test by exploiting two sources of post-treatment investment variation.

If the home-environment interaction β_5^N is zero but the school-quality interaction β_4^N is positive (or vice versa), this would suggest that complementarity operates on one margin but not the other—a result inconsistent with the CES technology (which implies a single ϕ governing the stock-investment interaction regardless of the source of investment) but potentially consistent with a model where different types of investment have different production functions. We note this possibility and do not claim the test covers all forms of complementarity.

As a further robustness check, we construct a composite investment index $\hat{I}_i$ by extracting the first principal component from school quality, HOME inventory, parenting warmth, and literacy frequency, following the measurement model approach of Cunha et al. (2010). The test $T_i \times \hat{I}_i$ on the composite is the closest analog to their structural specification.

6.4 Objection 3: “Three Years Is Not Thirty”

The HSIS follows children for approximately 4 years after treatment. Dynamic complementarity compounds: a small δ per period, accumulated over 30 years, produces a large cumulative interaction. The predicted β_3 at 3rd grade may be below the minimum detectable effect even if the long-run interaction is substantial.

Response. This is a legitimate concern, and the paper addresses it with a power calculation that uses Heckman’s own structural estimates to compute the predicted interaction at the HSIS horizon.

From Cunha et al. (2010), the noncognitive CES parameters at the early stage are $\hat{\phi}^N = -0.852$ and $\hat{\gamma}^N = 0.451$. Using these to compute the predicted treatment $\times$ investment interaction at the 3rd-grade follow-up (approximately 3 periods after treatment in the CHS two-stage framework):

The predicted β_3^N at period $n = 3$ under the CHS parameters, for a treatment effect of $\Delta\theta_0 = 0.2$ SD (the approximate end-of-Head-Start noncognitive effect from Puma et al.

2012) and a 1-SD variation in school quality, is:

$$\beta_3^{N,\text{pred}} \approx n \cdot \frac{\partial^2 f^N}{\partial \theta \partial I} \Big|_{\bar{\theta}, \bar{I}} \cdot \Delta \theta_0 \cdot \sigma_Q. \quad (42)$$

Evaluating the cross-partial (3) at the CHS parameter values and at the sample means of θ and I (normalized to 1) gives $\partial^2 f^N / \partial \theta \partial I \approx 0.45 \cdot 0.55 \cdot 1.852 \cdot 1^{-1.852} \cdot 1^{-1.852} \cdot 1^{-1.704} \approx 0.46$. Therefore:

$$\beta_3^{N,\text{pred}} \approx 3 \times 0.46 \times 0.2 \times 1.0 \approx 0.28 \text{ SD}. \quad (43)$$

This is a large effect—well above the minimum detectable effect at the HSIS sample size ($n \approx 5,000$; MDE at 80% power ≈ 0.08 SD for the interaction term). If the CHS structural parameters are correct, the HSIS has ample power to detect complementarity at the 3rd-grade horizon. A null finding at $n = 5,000$ with MDE = 0.08 SD would reject the CHS noncognitive complementarity estimates, not merely fail to detect a small effect.

We report this power calculation in the results to discipline the interpretation: a null finding is informative (it rejects the magnitude of complementarity implied by CHS) rather than merely inconclusive.

If the predicted β_3^N under CHS parameters turns out to be below the MDE—which we compute formally before looking at the data—we state this explicitly: the test has low power against the CHS alternative at this horizon, and a null finding would be uninformative. In that case, the paper shifts to the auxiliary tests (wealth, mediation, trajectories, gender), which use Perry and Abecedarian data at 20–40 year horizons where complementarity has had time to compound. The weakness of the 3rd-grade horizon is offset by the strength of the long-run auxiliary evidence. No single test settles the question; the five tests jointly do.

6.5 Objection 4: “Your Noncognitive Measures Are Noisy”

Teacher-reported behavioral scales are subjective, noisy, and correlated with teacher characteristics. Measurement error in y^N attenuates the treatment effect and inflates the standard error on β_3^N , biasing the test toward the null. Failure to reject $\beta_3^N = 0$ may reflect noise, not invariance.

Response. Three approaches mitigate this.

First, following Heckman et al. (2013), we extract a latent noncognitive factor from the

multiple behavioral subscales using a measurement model:

$$y_i^{N,j} = \lambda_j \theta_i^N + v_i^j, \quad j = 1, \dots, J, \quad (44)$$

where j indexes the behavioral subscales, θ_i^N is the latent factor, and v_i^j is subscale-specific measurement error. The factor $\hat{\theta}_i^N$ is estimated by maximum likelihood or by the weighted average $\hat{\theta}_i^N = (\sum_j \hat{\lambda}_j y_i^{N,j}) / (\sum_j \hat{\lambda}_j^2)$, which is the minimum-variance linear combination. The test is then $T_i \times Q_i$ on $\hat{\theta}_i^N$, which has higher signal-to-noise than any individual subscale.

Second, we use *behavioral* outcomes downstream of noncognitive traits: grade retention, special education placement, and disciplinary incidents. These are recorded in administrative data, not teacher reports, and are binary or count variables with no subjective rating scale. They are noisier measures of the latent trait (further downstream in the causal chain) but are free of reporter bias.

Third, we bound the attenuation. If the reliability of the noncognitive measure is λ^2 , the attenuated interaction coefficient is $\hat{\beta}_3^N = \lambda^2 \beta_3^{N,\text{true}}$. Under the CHS structural parameters, $\beta_3^{N,\text{true}} \approx 0.28$ SD (from the power calculation above). Even with reliability as low as $\lambda^2 = 0.5$ (plausible for teacher reports), the attenuated coefficient $\hat{\beta}_3^N \approx 0.14$ SD is still above the MDE of 0.08. The attenuation makes the test harder to pass, but not impossible if the true interaction is as large as CHS implies. The paper reports the reliability-adjusted MDE alongside the raw MDE.

6.6 Objection 5: “The Dual-Process Hypothesis Is Unfalsifiable”

Under the dual-process prediction ($\delta^C > 0, \delta^N = 0$), a finding of $\beta_3^C > 0, \beta_3^N = 0$ confirms the paper; $\beta_3^C > 0, \beta_3^N > 0$ confirms skills; and $\beta_3^C = 0, \beta_3^N = 0$ confirms formation. What result would reject the paper?

Response. Two results would reject the formation model decisively:

- (a) $\hat{\beta}_3^N > 0$ at the HSIS horizon, with magnitude consistent with the CHS structural parameters (i.e., $\hat{\beta}_3^N \geq 0.14$ SD after reliability adjustment). This would mean noncognitive traits interact with post-treatment investment, exactly as the skills model predicts and the formation model denies.
- (b) $\hat{\beta}_3^N > 0$ in the translog specification (40), which is robust to the linearization critique. If the translog interaction $\hat{\alpha}_3^N > 0$, the result holds regardless of functional form.

The paper pre-commits to these falsification criteria. Specifically: if $\hat{\beta}_3^N$ exceeds the reliability-adjusted MDE (0.14 SD under CHS parameters with $\lambda^2 = 0.5$, or 0.08 SD with $\lambda^2 = 1$) and is statistically significant at the 5% level in at least two of the three specifications (linear, CES, translog), the formation model is rejected for the noncognitive dimension. The paper reports the exact threshold before examining the data.

Additionally, the five predictions are jointly testable. If two or more predictions go against the formation model—for instance, if both the complementarity test and the mediation test favor skills ($\hat{\beta}_3^N > 0$ and within-occupation direct effects are significant)—the weight of evidence shifts decisively toward skills. No single test is conclusive; the inferential logic is Bayesian over the five tests.

6.7 Objection 6: “You Are Relabeling My Self-Productivity as Your Cultural Persistence”

In the CHS estimates, noncognitive self-productivity ($\partial f^N / \partial \theta_t^N$) reaches 0.662 at the later stage. The formation model’s ρ is also an autoregressive coefficient on θ_t . The two are observationally equivalent in the first-order dynamics. Calling the autoregressive component “cultural persistence” rather than “self-productivity” is a relabeling, not a discovery.

Response. This is correct at the level of the univariate time series. The autoregressive coefficient ρ in the formation model and the self-productivity parameter f_θ in the CES generate identical dynamics when $\delta = 0$. If the test finds $\hat{\delta}^N = 0$, the data cannot distinguish the two interpretations of the autoregressive component.

The paper’s claim is therefore precise: if $\hat{\delta}^N = 0$, then noncognitive persistence is *consistent with* both self-productivity and cultural transmission, but the *mechanism* through which the autoregressive component operates is not identified by the interaction test alone. The paper argues for the formation interpretation on three grounds that are external to the δ test:

1. *Microfoundational parsimony*: three distinct mechanisms (Bayesian updating, Bisin-Verdier cultural transmission, BBMZ identity) independently generate additive persistence from deeper primitives. Self-productivity in the CES is a primitive—it does not decompose further.
2. *Wealth excess* (Prediction 2): if the autoregressive component is self-productivity (a production function property), it does not change the saving rate. If it is cultural persistence of a preference (formation), it changes both behavior and the saving rate. The wealth-to-earnings multiplier $\Delta > 1$ observed in Perry is consistent with

formation and inconsistent with self-productivity alone.

3. *Mediation pattern* (Prediction 3): self-productivity in the CES implies that the skill stock raises within-occupation productivity. Cultural persistence of a preference implies the stock changes occupational choice but not within-occupation productivity. The Heckman-Pinto-Savelyev decomposition shows the latter.

The honest statement is: the δ test distinguishes complementarity from non-complementarity. Conditional on $\delta^N = 0$, the auxiliary tests (wealth, mediation) distinguish self-productivity from preference persistence. No single test does all the work. The joint evidence— $\hat{\delta}^N = 0$ plus $\Delta > 1$ plus within-occupation direct effect ≈ 0 —is the basis for concluding formation rather than self-productivity.

6.8 Joint Structural Estimation

Beyond the reduced-form tests, we estimate the nested model (17) structurally via method of simulated moments, matching five moment groups:

1. Noncognitive treatment effects at end-of-HS, 1st grade, 3rd grade (identifies $\rho + \delta\bar{I}$).
2. Treatment $\times$ school-quality interaction at each wave, separately for cognitive and noncognitive outcomes (identifies δ^C and δ^N).
3. Cognitive treatment effect trajectory: end-of-HS, 1st grade, 3rd grade (identifies the cognitive process—stock decay vs. flow removal).
4. Home environment $\times$ treatment interaction (identifies whether complementarity operates through parental or school investment).
5. Pre-treatment SES $\times$ treatment interaction (formation predicts larger effects for more disadvantaged children, because the signal contrast $\Delta\lambda$ is larger; skills predicts effects that may be larger or smaller depending on the concavity of the production function at different stock levels).

The structural parameters are $\Theta = (\rho^N, \delta^N, \rho^C, \delta^C, \mu^N, \mu^C)$ —six parameters from five moment groups plus the trajectory shape, which provides over-identifying restrictions. The simulated model is the dual-process system (24)–(25), iterated forward from the end-of-Head-Start baseline for each child, with the child’s actual post-treatment school quality and home environment as inputs.

Standard errors are computed by bootstrap (resampling children within centers to preserve the clustered randomization design). The over-identification test compares the minimized objective to a χ^2 distribution with degrees of freedom equal to the number of moments minus the number of parameters.

7 Threats to Identification

Endogenous school choice. If treated children sort into better schools because higher θ leads to better decisions, the treatment $\times$ quality interaction is confounded: $\beta_3 > 0$ could reflect selection rather than complementarity. The HSIS randomization ensures $T_i \perp Q_i | X_i$ at baseline, but if treatment changes school choice (an outcome of the intervention), post-treatment school quality is endogenous.

We address this with an IV approach: instrument $Q_{i,\text{post}}$ with the *mean school quality in the child's baseline residential census tract*, which captures the quality of the default school the child would have attended absent any treatment-induced sorting. The exclusion restriction is that tract-level school quality affects child outcomes only through the school actually attended, conditional on baseline demographics. We also report the reduced-form specification with tract-level quality directly (replacing $Q_{i,\text{post}}$ with Q_{tract}), which is immune to sorting since tract is determined before treatment.

Community-level signals vs. individual-level complementarity. The Johnson-Jackson interaction may reflect a richer community-level signal environment rather than individual-level skill-investment complementarity. Counties with both Head Start and high school spending may deposit higher θ through the formation channel's $\mu(\lambda r - r^*)$ term. The HSIS design addresses this: within the same county (and often within the same center catchment area), children attend different post-treatment schools of varying quality. The within-county, within-center variation in school quality is driven by residential location and school assignment boundaries, not by county-level policy. The treatment $\times$ school-quality interaction estimated from this variation isolates individual-level complementarity from community-level signals.

State-dependent identity. Under the BBMZ extension with state-dependent β —where agents with higher θ resist downward revision more strongly—treated children in bad post-treatment environments may *defend* their higher θ against the negative signal, generating $\beta_3^N < 0$: the treatment effect is *larger* in bad environments, not smaller. This is Outcome D (the negative-interaction case). It would reject both the baseline formation model and the skills model. We test for this explicitly by allowing the interaction to be negative and reporting a two-sided test.

Biological embedding. Campbell et al. (2014) found that Abecedarian treatment reduced male metabolic syndrome prevalence from 25% to 0% at age 35. If early environ-

ments become biologically embedded through epigenetic mechanisms (Hertzman and Boyce, 2010), persistence may operate through a physiological channel outside both the skill and preference frameworks. We cannot rule this out, and acknowledge that a portion of the “ ρ ” we estimate may reflect biological persistence rather than cultural transmission. The biological channel is observationally equivalent to formation in the δ test (both predict $\delta^N = 0$) but is distinguishable in principle through biomarker data (cortisol, epigenetic markers) that the HSIS does not contain. This is a limitation.

Multiple testing. Five predictions tested simultaneously inflate the family-wise error rate. We adjust using the Benjamini-Hochberg procedure for false discovery rate control at 5%, applied across the five primary tests. The pre-committed falsification criterion ($\hat{\beta}_3^N > 0.14$ SD in two of three specifications) is assessed at the individual-test level; the joint assessment across all five predictions uses the FDR-adjusted p -values.

8 Optimal Early Childhood Investment under Skills vs. Formation

The two models imply different optimal allocations of a fixed early childhood budget. This section derives the optimal investment schedule under each model and shows that the welfare cost of misallocation—applying the skills-optimal policy when the formation model is correct, or vice versa—is large.

8.1 Setup

A planner has budget B to allocate across T years of a child’s life ($t = 0, 1, \dots, T - 1$, where $t = 0$ is birth and $T = 18$ is the end of childhood). The allocation $\{I_t\}$ must satisfy $\sum_t I_t \leq B$. The planner’s objective is to maximize the child’s noncognitive stock at adulthood, θ_T .

Under the nested model, the law of motion is:

$$\theta_{t+1} = (\rho + \delta I_t) \theta_t + (1 - \rho) \bar{\theta} + \mu I_t, \quad (45)$$

where we simplify by treating I_t as the investment that enters both the direct adaptation channel (through μI_t , replacing $\mu(\lambda r_t - r^*)$ for notational clarity) and the complementarity channel (through $\delta \theta_t I_t$). The signal contrast $\lambda_{\text{treat}} - \lambda_{\text{control}}$ is absorbed into I_t .

8.2 The Skills-Optimal Allocation ($\delta > 0, \rho = 0$)

When persistence operates entirely through complementarity, early investment compounds: a dollar at $t = 0$ interacts with $T - 1$ subsequent periods of investment, while a dollar at $t = T - 1$ interacts with none. The return to investment at period t includes both the direct effect (μ) and the compounded complementarity effect ($\delta\theta_t$ applied to all future investments).

The Lagrangian gives the FOC:

$$\mu + \delta\theta_t + \delta \sum_{s=t+1}^{T-1} \frac{\partial\theta_T}{\partial\theta_s} \cdot \theta_s = \nu \quad \forall t, \quad (46)$$

where ν is the shadow value of the budget constraint. Since θ_t is higher at later periods (due to accumulation) but the remaining compounding horizon $T - t$ is shorter, the FOC generically implies a *front-loaded but sustained* allocation: the planner invests most at $t = 0$ (when compounding time is longest) but continues investing throughout childhood because each subsequent period's investment is amplified by the accumulated stock. The optimal allocation tapers: $I_0^* > I_1^* > \dots > I_{T-1}^* > 0$ when $\delta > 0$.

Under the CHS estimates ($\hat{\phi}^N = -0.852$ in early childhood), the complementarity is strong and the front-loading is steep. But because $\delta > 0$, the planner *never* sets $I_t = 0$ for any t —zero investment at any period wastes the accumulated stock.

8.3 The Formation-Optimal Allocation ($\delta = 0, \rho > 0$)

When persistence operates through cultural transmission and the cross-partial is zero, investment in period t produces a direct effect μ on θ_{t+1} that persists at rate ρ through cultural channels. Investment in period $s > t$ does *not* interact with the accumulated stock—each dollar operates independently.

The return to a dollar invested at period t is:

$$\frac{\partial\theta_T}{\partial I_t} = \mu \rho^{T-t-1}. \quad (47)$$

This is a *monotonically decreasing* function of t : earlier investment persists longer (ρ^{T-t-1} is higher when t is small). The key difference from the skills model: the return depends *only* on the remaining persistence horizon, not on the accumulated stock. There is no interaction with subsequent investment.

The optimal allocation under formation is a *corner solution*: the planner concentrates

the entire budget in the earliest feasible period:

$$I_0^* = B, \quad I_t^* = 0 \quad \text{for } t \geq 1. \quad (48)$$

No investment after the initial period is warranted, because later investment does not compound through the stock—it just persists at rate ρ , which is lower from a later starting point. The marginal return to investment is strictly decreasing in t and does not depend on how much was invested before.

In practice, the formation window is not a single period but spans approximately ages 0–5 (the period during which the Kalman gain κ_t is high and preferences are still plastic). The optimal policy concentrates investment within this window—not at a single instant—but *zero* investment is optimal after the window closes. This is a bang-bang solution: full intensity during the formation window, zero after. Contrast with the skills-optimal tapering: positive investment throughout, declining but never zero.

8.4 The Formation Share Determines the Optimal Horizon

Under the general nested model with $\rho > 0$ and $\delta > 0$, the formation share ω_F (equation 27) determines the optimal investment horizon.

Proposition 3 (Optimal Horizon). *Let T^* be the number of periods in which optimal investment is strictly positive. Then:*

- (a) *When $\omega_F = 1$ ($\delta = 0$): $T^* = A$, the length of the formation window. All investment occurs during ages 0– A .*
- (b) *When $\omega_F = 0$ ($\rho = 0$): $T^* = T$. Investment is positive throughout childhood.*
- (c) *For intermediate ω_F : T^* is decreasing in ω_F . Higher formation share $\Rightarrow$ shorter optimal investment horizon.*

Proof sketch. The marginal return to investment at period t in the nested model is:

$$\frac{\partial \theta_T}{\partial I_t} = \mu \prod_{s=t+1}^{T-1} (\rho + \delta I_s) + \delta \theta_t \prod_{s=t+1}^{T-1} (\rho + \delta I_s). \quad (49)$$

The first term (direct signal effect) is present under both models; the second (complementarity) is present only when $\delta > 0$. When $\delta = 0$, the return at t is $\mu \rho^{T-t-1}$, which is maximized at $t = 0$ and decreasing. The planner exhausts the budget as early as possible. When $\delta > 0$, the complementarity term $\delta \theta_t$ grows with the accumulated stock, partially

offsetting the decline in the persistence horizon. The higher δ relative to ρ , the more the complementarity offset sustains positive investment in later periods. $\square$

8.5 The Welfare Cost of Misallocation

The practical stake: what happens if the planner applies the wrong model?

Scenario 1: Formation is correct but the planner applies the skills model. The planner tapers investment across 18 years instead of concentrating in the formation window. The loss comes from two sources: (a) investment after the formation window ($t > A$) has no signal effect ($\mu = 0$ when the window is closed) and no complementarity effect ($\delta^N = 0$), so it is wasted; (b) investment during the formation window is diluted (the budget is spread across 18 years rather than concentrated in 5). For a budget B spread uniformly over 18 years, the investment during the formation window is $B \cdot 5/18 \approx 0.28B$, versus B under the formation-optimal allocation. The loss in $\Delta\theta_0$ is approximately $(1 - 5/18) \approx 72\%$ of the achievable preference deposit.

Scenario 2: Skills is correct but the planner applies the formation model. The planner concentrates the entire budget in ages 0–5 and invests zero in ages 6–18. The loss comes from the foregone complementarity: the initial skill stock θ_5 depreciates without subsequent investment to sustain it. Under the CHS noncognitive parameters, the treatment effect at age 18 is:

$$D_{18} = D_5 \cdot \prod_{t=6}^{17} (\rho + \delta I_t). \quad (50)$$

With $I_t = 0$ for $t > 5$: $D_{18} = D_5 \cdot \rho^{12}$. With optimal sustained investment: $D_{18} = D_5 \cdot (\rho + \delta \bar{I})^{12} > D_5 \rho^{12}$. The ratio of the misallocated to the correctly-allocated effect is $[\rho / (\rho + \delta \bar{I})]^{12} = \omega_F^{12}$. For $\omega_F = 0.5$: $0.5^{12} \approx 0.0002$ —essentially zero. The formation-optimal policy is catastrophic when the skills model is correct and ω_F is low.

The asymmetry is revealing. If formation is correct and the planner applies the skills model, 72% of the budget is wasted but the remaining 28% still produces a deposit. If skills is correct and the planner applies the formation model, the initial deposit decays to near zero without sustained investment. *The skills-model error under formation dominance is wasteful; the formation-model error under skills dominance is fatal.* This asymmetry favors a precautionary strategy: when uncertain about ω_F , the planner should invest more during the formation window (insuring against formation dominance) while maintaining some

sustained investment (insuring against skills dominance). The optimal precautionary allocation depends on the prior over ω_F and the relative magnitudes of the two losses.

Remark 4. *The welfare analysis highlights why the δ test matters practically. Current early childhood policy in the United States (Head Start, Pre-K programs) implicitly assumes the skills model: invest in structured instruction during the preschool years, then hand the child to the regular school system. The formation model implies a radically different design: invest in the signal environment—the adults, the role models, the visible futures—during the formation window, and accept that the regular school system’s contribution to noncognitive development is maintenance, not complementation. The difference is whether we build a better classroom or a different world for the child to grow up in. At a federal Head Start budget of approximately \$12 billion per year, the misallocation cost of applying the wrong model is measured in billions.*

9 Conclusion

The technology of skill formation and the adaptive preference formation model are nested by a single parameter δ : the cross-partial of the transition function with respect to the current non-cognitive stock and subsequent investment. Under skills, $\delta > 0$. Under formation, $\delta = 0$.

The existing evidence—drawn from five published tests across four landmark experiments—jointly favors a dual-process interpretation. Cognitive outcomes follow the skills model: they respond to current instructional flow, interact with post-treatment quality, and fade when the flow is removed. Non-cognitive outcomes follow the formation model: the treatment changed the developmental signal environment, deposited a preference (θ) that persists at rate ρ regardless of the subsequent environment, and manifests through accumulated behavioral choices that compound over decades.

The practical stake is immediate. The Cunha-Heckman framework implies: invest early *and sustain* the investment—sequence cognitive and non-cognitive inputs to exploit complementarity over 18 years. The formation framework implies: *change the park* during the formation window—invest intensively in the signal environment (the teachers, the role models, the visible futures) and let the preference deposit persist on its own. At any given budget, the two policies produce very different programs.

The RCT evidence to settle the question exists. The HSIS microdata (ICPSR Study 36968) contain random treatment assignment, detailed non-cognitive outcome measures, and post-treatment school identifiers. What has not been done—and what this paper proposes—is to estimate the treatment $\times$ school-quality interaction *separately for cognitive*

and non-cognitive outcomes within a structural model that nests both mechanisms. If the interaction is zero for noncognitive outcomes and positive for cognitive outcomes, both Heckman and this paper are right—but about different things. Skills for cognition. Parks for preferences. And it is the park that persists.

References

- Agostinelli, F. and M. Wiswall. 2025. "Estimating the Technology of Children's Skill Formation." *Journal of Political Economy* 133(3).
- Bailey, D., G.J. Duncan, C.L. Odgers, and W. Yu. 2017. "Persistence and Fadeout in the Impacts of Child and Adolescent Interventions." *Journal of Research on Educational Effectiveness* 10(1): 7–39.
- Bailey, D., G.J. Duncan, F. Cunha, B.R. Foorman, and D.S. Yeager. 2020. "Persistence and Fade-Out of Educational-Intervention Effects." *Psychological Science in the Public Interest* 21(2): 55–97.
- Becker, G.S. and C.B. Mulligan. 1997. "The Endogenous Determination of Time Preference." *Quarterly Journal of Economics* 112(3): 729–758.
- Bisin, A. and T. Verdier. 2001. "The Economics of Cultural Transmission and the Dynamics of Preferences." *Journal of Economic Theory* 97(2): 298–319.
- Bisin, A. and T. Verdier. 2023. "Advances in the Economic Theory of Cultural Transmission." *Annual Review of Economics* 15: 63–89.
- Bowles, S. 1998. "Endogenous Preferences: The Cultural Consequences of Markets and Other Economic Institutions." *Journal of Economic Literature* 36(1): 75–111.
- Campbell, F.A., E.P. Pungello, K. Kainz, M. Burchinal, Y. Pan, B.H. Wasik, O. Barbarin, J.J. Sparling, and C.T. Ramey. 2012. "Adult Outcomes as a Function of an Early Childhood Educational Program: An Abecedarian Project Follow-Up." *Developmental Psychology* 48(4): 1033–1043.
- Campbell, F., G. Conti, J.J. Heckman, S.H. Moon, R. Pinto, E. Pungello, and Y. Pan. 2014. "Early Childhood Investments Substantially Boost Adult Health." *Science* 343(6178): 1478–1485.
- Cappelen, A.W., J.A. List, A. Samek, and B. Tungodden. 2020. "The Effect of Early-Childhood Education on Social Preferences." *Journal of Political Economy* 128(7): 2739–2758.
- Castillo, M., J.A. List, R.A. Petrie, and A. Samek. 2024. "Detecting Drivers of Behavior at an Early Age: Evidence from a Longitudinal Field Experiment." *Journal of Political Economy* 132(12): 3942–3977.

- Cunha, F. and J.J. Heckman. 2007. "The Technology of Skill Formation." *American Economic Review P&P* 97(2): 31–47.
- Cunha, F., J.J. Heckman, and S.M. Schennach. 2010. "Estimating the Technology of Cognitive and Noncognitive Skill Formation." *Econometrica* 78(3): 883–931.
- Deming, D. 2009. "Early Childhood Intervention and Life-Cycle Skill Development: Evidence from Head Start." *American Economic Journal: Applied Economics* 1(3): 111–134.
- Doepke, M. and F. Zilibotti. 2017. "Parenting with Style: Altruism and Paternalism in Intergenerational Preference Transmission." *Econometrica* 85(5): 1331–1371.
- García, J.L., J.J. Heckman, D.E. Leaf, and M.J. Prados. 2020. "Quantifying the Life-Cycle Benefits of an Influential Early-Childhood Program." *Journal of Political Economy* 128(7): 2502–2541.
- García, J.L., J.J. Heckman, and V. Ronda. 2023. "The Lasting Effects of Early-Childhood Education on Promoting the Skills and Social Mobility of Disadvantaged African Americans and Their Children." *Journal of Political Economy* 131(6): 1477–1506.
- Gertler, P., J. Heckman, R. Pinto, A. Zanolini, C. Vermeersch, S. Walker, S.M. Chang, and S. Grantham-McGregor. 2014. "Labor Market Returns to an Early Childhood Stimulation Intervention in Jamaica." *Science* 344(6187): 998–1001.
- Gertler, P., J.J. Heckman, R. Pinto, S. Chang, S. Grantham-McGregor, C. Vermeersch, S. Walker, and A. Wright. 2021. "Effect of the Jamaica Early Childhood Stimulation Intervention on Labor Market Outcomes at Age 31." NBER Working Paper No. 29292.
- Goff, L., O. Malamud, C. Pop-Eleches, and M. Urquiola. 2025. "Interactions Between Family and School Environments." *Journal of Human Resources* 60(3): 907–949.
- Hart, E.R., D.H. Bailey, Y. Luo, K. Sengupta, and T.W. Watts. 2024. "Fadeout and Persistence of Intervention Impacts on Social-Emotional and Cognitive Skills in Children and Adolescents: A Meta-Analytic Review." *Psychological Bulletin* 150(10): 1207–1236.
- Heckman, J.J., S.H. Moon, R. Pinto, P. Savelyev, and A. Yavitz. 2010. "The Rate of Return to the HighScope Perry Preschool Program." *Journal of Public Economics* 94(1–2): 114–128.
- Heckman, J.J. and S. Mosso. 2014. "The Economics of Human Development and Social Mobility." *Annual Review of Economics* 6: 689–733.

- Heckman, J.J., R. Pinto, and P. Savelyev. 2013. "Understanding the Mechanisms Through Which an Influential Early Childhood Program Boosted Adult Outcomes." *American Economic Review* 103(6): 2052–2086.
- Hertzman, C. and T. Boyce. 2010. "How Experience Gets Under the Skin to Create Gradients in Developmental Health." *Annual Review of Public Health* 31: 329–347.
- Johnson, R.C. and C.K. Jackson. 2019. "Reducing Inequality through Dynamic Complementarity: Evidence from Head Start and Public School Spending." *American Economic Journal: Economic Policy* 11(4): 310–349.
- Kline, P. and C.R. Walters. 2016. "Evaluating Public Programs with Close Substitutes: The Case of Head Start." *Quarterly Journal of Economics* 131(4): 1795–1848.
- Protzko, J. 2015. "The Environment in Raising Early Intelligence: A Meta-Analysis of the Fadeout Effect." *Intelligence* 53: 202–210.
- Puma, M., S. Bell, R. Cook, C. Heid, P. Broene, F. Jenkins, A. Mashburn, and J. Downer. 2012. "Third Grade Follow-up to the Head Start Impact Study." OPRE Report 2012-45.
- Rossin-Slater, M. and M. Wüst. 2020. "What Is the Added Value of Preschool? Long-Term Impacts and Interactions with an Infant Health Intervention." *American Economic Journal: Applied Economics* 12(3): 255–286.
- Schiman, J.C. 2024. "Do Preschool Investments Depreciate and Fadeout or Result in Dynamic Complementarities? An Assessment Using the Head Start Impact Study." *Review of Economics of the Household* 22: 1181–1211.
- Schweinhart, L.J., J. Montie, Z. Xiang, W.S. Barnett, C.R. Belfield, and M. Nores. 2005. *Lifetime Effects: The High/Scope Perry Preschool Study Through Age 40*. High/Scope Press.
- Sen, A. 1985. *Commodities and Capabilities*. North-Holland.
- Walker, S.P., S.M. Chang, M. Vera-Hernández, and S. Grantham-McGregor. 2022. "Cognitive, Psychosocial, and Behaviour Gains at Age 31 Years from the Jamaica Early Childhood Stimulation Trial." *Journal of Child Psychology and Psychiatry* 63(6): 626–635.